

Generation Interconnection System Impact Study Report

For Queue Project Q15-0461

Anthony 138 kV
September 21, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0461. The Interconnection Customer proposes a 334.6 MW wind facility interconnecting at the Anthony 138 kV substation in the Virginia area. The request was received on April 14, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 238 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,367,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Anthony to Slabtown 138 kV
Contingency: Anthony to Dunn Siding 138 kV Line
Loading: 153.6% (Rating: 201 MVA)
DFAX: 100.0%
Pre-project loading: 13.0%
2. Facility: Dunn Siding to Anthony 138 kV
Contingency: Anthony to Slabtown 138 kV
Loading: 136.3% (Rating: 226 MVA)
DFAX: 100.0%
Pre-project loading: 11.5%
3. Facility: Slabtown to Skaggs Branch Substation 138 kV line
Contingency: Tuckasegee Tie to Mered 138 kV
Loading: 129.8% (Rating: 242 MVA)
DFAX: 26.8%
Pre-project loading: 92.6%
4. Facility: Contra Costa to Libby 138 kV Line
Contingency: Contra Costa to Birchwood 138 kV Line
Loading: 129.7% (Rating: 439 MVA)
DFAX: 26.6%
Pre-project loading: 109.4%
5. Facility: Skaggs Branch to Sperry 138 kV
Contingency: Merced Substation to Tuckasegee Tie Substation 138 kV Line
Loading: 123.0% (Rating: 232 MVA)
DFAX: 26.8%
Pre-project loading: 84.4%
6. Facility: Contra Costa to Miller 138 kV Ckt 1
Contingency: Contra osta to Miller 138 kV
Loading: 121.6% (Rating: 495 MVA)

DFAX: 19.8%

Pre-project loading: 108.2%

7. Facility: Miller to Contra Costa 138 kV Line 2
Contingency: Miller to Contr Costa 138 kV Ckt 1
Loading: 121.6% (Rating: 495 MVA)
DFAX: 19.8%
Pre-project loading: 108.2%
8. Facility: Lookout Substation to Nathan 138 kV
Contingency: Libby to Contra Costa 138 kV
Loading: 120.9% (Rating: 231 MVA)
DFAX: 6.1%
Pre-project loading: 112.0%
9. Facility: Ladysmith t to Dunn Siding 138 kV
Contingency: Libby to Contra Costa 138 kV
Loading: 119.5% (Rating: 231 MVA)
DFAX: 33.7%
Pre-project loading: 70.6%
10. Facility: Sperry to Libby 138 kV L
Contingency: Tuckasegee Tie to Merced 138 kV
Loading: 113.9% (Rating: 223 MVA)
DFAX: 26.8%
Pre-project loading: 73.7%
11. Facility: Adrian Energy Associates Llc 345 kV/138 kV XFMR
Contingency: Taber to Glenannan 345 kV
Loading: 112.2% (Rating: 750 MVA)
DFAX: 5.0%
Pre-project loading: 109.9%
12. Facility: Highgrove to Ctr 138 kV
Contingency: Dixleta to Taber Substation 345 kV Line
Loading: 108.1% (Rating: 417 MVA)
DFAX: 5.2%
Pre-project loading: 103.9%
13. Facility: Bells Mill to Beverly Farms 138 kV
Contingency: Tiger Bay to Dranesville 345 kV line
Loading: 105.8% (Rating: 413 MVA)
DFAX: 6.5%
Pre-project loading: 100.6%
14. Facility: City W Diesel Plant to Nauvoo 345 kV
Contingency: Nauvoo to Roberts Sw Sta 345 kV
Loading: 105.0% (Rating: 1,338 MVA)
DFAX: 7.6%
Pre-project loading: 103.0%
15. Facility: Rossmoor to Miller 138 kV
Contingency: Miller to Mullin 138 kV
Loading: 104.5% (Rating: 243 MVA)
DFAX: 5.9%
Pre-project loading: 96.4%
16. Facility: Dunn Siding to Fredericksburg 345 kV
Contingency: Nauvoo to Birchwood 345 kV
Loading: 104.3% (Rating: 1,596 MVA)
DFAX: 16.7%

Pre-project loading: 100.8%

17. Facility: Glenannan to Roberts Sw Sta 345 kV

Contingency: Glenannan to Cal Sierra Development 345 kV Line

Loading: 103.9% (Rating: 1,223 MVA)

DFAX: 10.2%

Pre-project loading: 101.1%

18. Facility: Nauvoo to Birchwood 345 kV

Contingency: Fredericksburg to Dunn Siding 345 kV

Loading: 101.0% (Rating: 2,104 MVA)

DFAX: 13.1%

Pre-project loading: 98.9%

19. Facility: Miller to Mullin 138 kV

Contingency: Miller to Rossmoor 138 kV

Loading: 100.5% (Rating: 238 MVA)

DFAX: 5.6%

Pre-project loading: 92.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Anthony to Slabtown 138 kV	Rebuild line with higher-capacity conductor	\$3,779,000	100% (\$3,779,000)
Dunn Siding to Anthony 138 kV	Rebuild line with higher-capacity conductor	\$10,244,000	100% (\$10,244,000)
Skaggs Branch to Slabtown 138 kV	Rebuild line with higher-capacity conductor	\$5,638,000	100% (\$5,638,000)
Contra Costa to Libby 138 kV	Rebuild line with higher-capacity conductor	\$3,255,000	40% (\$1,312,000)
Skaggs Branch to Sperry 138 kV	Rebuild line with higher-capacity conductor	\$14,932,000	100% (\$14,932,000)
Miller to Contra Costa 138 kV Line 1	Rebuild line with higher-capacity conductor	\$18,214,000	37% (\$6,783,000)
Miller to Contra Costa 138 kV Line 2	Rebuild line with higher-capacity conductor	\$22,476,000	37% (\$8,371,000)
Lookout to Nathan 138 kV L	Rebuild line with higher-capacity conductor	\$13,365,000	100% (\$13,365,000)
Ladysmith t to Dunn Siding 138 kV	Rebuild line with higher-capacity conductor	\$4,219,000	60% (\$2,536,000)
Sperry to Libby 138 kV	Rebuild line with higher-capacity conductor	\$1,455,000	100% (\$1,455,000)
Adrian Energy Associates Llc 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$15,323,000	13% (\$1,947,000)
Center to Highgrove 138 kV	Reconductor line	\$1,457,000	4% (\$63,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bells Mill to Beverly Farms 138 kV	Reconductor line	\$806,000	3% (\$23,000)
City W Diesel Plant to Nauvoo 345 kV	Reconductor line	\$16,496,000	3% (\$560,000)
Rossmoor to Miller 138 kV	Reconductor line	\$567,000	100% (\$567,000)
Dunn Siding to Fredericksburg 345 kV	Reconductor line	\$17,605,000	6% (\$1,048,000)
Roberts Sw Sta to Glenannan 345 kV Line	Reconductor line	\$23,214,000	3% (\$624,000)
Nauvoo to Birchwood 345 kV	Reconductor line	\$29,733,000	3% (\$998,000)
Mullin Substation to Miller 138 kV	Reconductor line	\$1,122,000	100% (\$1,122,000)
	Total allocated to project		\$75,367,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.